
Project Proposal

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1 Introduction

Vitamin deficiencies are some of the most prevalent yet underdiagnosed conditions globally. Symptoms such as fatigue, hair loss, and muscle weakness are often dismissed, whereas blood tests are inaccessible or inconvenient for many patients. Early identification of deficiencies like anemia, scurvy, or rickets can prevent serious long-term health consequences, yet no accessible tool currently allows a person to describe their symptoms in plain language and receive both a likely diagnosis and a dietary intervention in one interaction.

This proposal presents the plan for a capstone project that addresses this gap directly. We develop a machine learning pipeline that classifies vitamin deficiency type from patient-reported symptoms and demographics, then passes the predicted diagnosis to a large language model to generate a personalized meal plan. This system will provide an easily accessible screening tool that could help individuals make informed dietary decisions.

What makes our approach unique is the combination of three elements that prior work has never combined: multi-class symptom-based deficiency classification without blood work, free-text natural language input parsed by an LLM, and an automatically generated dietary intervention tied to the predicted diagnosis. We review relevant literature below to establish where existing methods succeed, where they fall short, and how our approach builds on and extends them.

2 Related Work Survey

We reviewed multiple prior works relevant to our project domain. These papers represent different algorithmic approaches, theoretical perspectives, and evaluation strategies.

2.1 Large language models in clinical nutrition: an overview of its applications, capabilities, limitations, and potential future prospects (Belkhouribchia & Pen, 2025)

This paper examines a number of studies related to using LLMs in clinical nutrition and summarizes the findings and limitations. Importantly to our project, it covers many examples regarding the creation of food plans or personalized diets. This helps us to understand what methods do and don't produce accurate, readable outcomes. Even more specifically, there are a couple that focus on creating food plans for particular diagnoses, such as malnutrition, diabetes, or obesity. These provide information about methods for developing food plans centered around a diagnosis, like vitamin deficiencies. What these seem to lack, however, is allowing the patient to directly interface with the LLM to obtain a diagnosis, followed by the development of a personalized diet.

This paper really delves into the strengths and weaknesses of the use of LLMs and where they excel and fall short in being used for clinical nutrition. As such, a main strength is that it allows us to identify ways to improve the functionality of the LLM and understand some of the limitations. On the other hand, it does not cover our main problem of diagnosing vitamin deficiencies from provided symptoms.

2.2 Machine learning-based prediction of vitamin D deficiency: NHANES 2001–2018 (Guo, He, & Li, 2024)

This study uses six ML models (XGBoost, random forest, logistic regression, SVM, neural network, and GBM) to predict vitamin D deficiency from just nine easily collected interview variables—things like age, race, BMI, milk consumption, and diabetes status—across 62,919 NHANES participants. This is relevant to our project because it demonstrates the feasibility of predicting a vitamin deficiency without requiring any blood work, using only demographic and lifestyle factors that a patient could self-report. The paper also deploys a practical Shiny web calculator and uses SHAP values to explain which features drive predictions, which aligns with the goal of building an accessible, interpretable tool.

A major strength is the massive, nationally representative dataset and the comprehensive metric reporting, including calibration (Brier score) and clinical utility (decision curve analysis). However, there is a critical weakness: the best model (XGBoost) reports a perfect AUC of 1.0 from nine simple interview questions, which is biologically implausible and strongly suggests data leakage or overfitting. Additionally, the study uses only a random train/test split with no external or temporal validation, race is the top predictor which raises fairness concerns, and season of blood draw—a known dominant driver of vitamin D levels—is entirely absent from the feature set.

2.3 Machine Learning-Driven Prediction of Vitamin D Deficiency Severity with Hybrid Optimization (Bhimavarapu, Battineni, & Chintalapudi, 2025)

This paper is quite similar to the previous one by Guo, He, and Li. Where it differs, however, is that this paper is focused on classifying the severity of the vitamin D deficiency (VDD). Based on 12 parameters collected from questionnaires, hospital records, and medical imaging, a dataset was preprocessed, and then the improved whale optimization (IWOA) algorithm was used to select 12 features for the severity prediction. Using 6 different machine learning models, the study evaluated key performance metrics such as precision, recall, F1-score, and accuracy. Ultimately, the models performed impressively well, with accuracy of up to 99.4%, as well as high precision, recall, and F1-scores. This suggests a strong case for using ML to accurately predict VDD severity. While our project is focused on the diagnoses and not necessarily the classification, it's still incredibly helpful to understand the methods used and the strengths and weaknesses of each. It's also encouraging to see that ML methods can very accurately predict medical conditions based on patient data.

2.4 Efficient Prediction of Vitamin B Deficiencies via Machine-Learning Using Routine Blood Test Results in Patients With Intense Psychiatric Episode (Tamune et al., 2020)

In this paper, they train random forest, SVM, logistic regression, and KNN models to predict vitamin B1, B12, and folate deficiency from 31 routine blood test parameters in 497 psychiatric emergency patients. This is important to our project because it is one of the very few papers that directly tackle B-vitamin deficiency prediction, and it honestly reveals a key challenge: folate deficiency was reasonably predictable (AUC 0.80), but B12 deficiency prediction was essentially no better than chance (AUC 0.60). This tells us that B12 deficiency may require different feature types—such as dietary history, medication use, or functional biomarkers like methylmalonic acid—rather than standard CBC parameters alone.

A notable strength is that it is the only paper in the broader literature that uses a temporal validation split (training on earlier months, testing on later months), which gives more realistic performance estimates than random splitting. It also tests multiple deficiency thresholds for robustness and provides interpretable feature importance. The main limitations are the very small sample size (only 13 B12-deficient patients in the validation set), the single-center all-Asian psychiatric population, which limits generalizability, and the fact that it does not allow a patient to interface directly with the model to obtain a diagnosis or personalized intervention plan.

2.5 Predictors of Micronutrient Deficiency Among Children Aged 6–23 Months in Ethiopia: A Machine Learning Approach (Gebeye LG, Dessie EY, & Yimam JA, 2024)

The paper applies five supervised ML models (Random Forest, SVM, Logistic Regression, Neural Network, and Naïve Bayes) to nationally representative 2019 EMDHS data to predict micronutrient

deficiency using demographic and socioeconomic variables rather than lab biomarkers. Random Forest performed best (AUROC 80.01 percent, accuracy 72.41 percent), identifying region, wealth index, maternal education, media exposure, and child age as key predictors. A major strength is the use of a large population level dataset and comparison across multiple models, along with spatial mapping and logistic regression benchmarking for interpretability. However, the outcome is based on a 24-hour dietary recall rather than biochemical confirmation, which weakens clinical validity, and the cross-sectional design limits causal inference. While performance is reasonable, it is not strong enough for diagnostic deployment, and deeper explainability methods are absent. Overall, the study supports the feasibility of non-invasive ML-based deficiency prediction but highlights the need for improved outcome measurement and validation.

2.6 Differential Diagnosis of Iron Deficiency Anemia from Aplastic Anemia Using Machine Learning and Explainable AI Utilizing Blood Attributes (Darshan BSD, Sampathila N, Bairy GM, Prabhu S, Belurkar S, Chadaga K, & Nandish S, 2025)

The paper develops ML models to distinguish iron deficiency anemia (IDA) from aplastic anemia (AA) using 23 routine blood parameters from a 500-patient retrospective dataset. Multiple classifiers and ensemble models were evaluated, with LightGBM and stacked models achieving high accuracy (about 96 percent), and five explainable AI techniques (SHAP, LIME, Eli5, Qlattice, Anchor) were used to interpret predictions. A major strength is its strong emphasis on explainability and clinical interpretability, making the model more trustworthy for decision support. It also demonstrates how deficiency-related conditions can be predicted from standard laboratory inputs. However, the study is limited by its small, single-center dataset and lack of external validation, which raises concerns about generalization. Unlike survey-based or symptom-based models in other papers, this study relies entirely on blood biomarkers, making it more clinically realistic but less applicable for early, non-invasive screening. Together with the other papers, it adds a strong explainable AI and differential diagnosis dimension while highlighting the need for broader validation and integration with earlier-stage diagnostic approaches.

2.7 Strengths and Weaknesses of Prior Work

The four papers share a consistent pattern, which is that existing ML approaches to vitamin deficiency prediction are either narrowly focused on a single vitamin, require clinical data like blood draws that patients cannot self-report (Tamune et al.), or evaluate prediction without connecting the diagnosis to any actionable intervention. The two vitamin D studies (Guo et al.; Bhimavarapu et al.) demonstrate that high classification accuracy is achievable from self-reported data alone, but neither allows a patient to interface directly with the model using free-text symptom descriptions. The LLM survey (Belkhouribchia & Pen) shows that language models can generate personalized dietary plans from a known diagnosis, but no reviewed work combines symptom-based diagnosis with LLM-driven meal planning in a single end-to-end pipeline.

Our work directly addresses these gaps in three ways. First, we extend the scope from single-vitamin prediction to multi-class classification across six deficiency categories simultaneously. Second, we replace structured clinical inputs with free-text symptom descriptions parsed by an LLM, making the tool accessible without any medical testing. Third, we close the loop that prior work leaves open by feeding the ML prediction directly into an LLM to generate a personalized meal plan, something none of the reviewed papers attempted.

3 Problem Definition

3.1 Lay-person Description

Vitamin deficiencies often cause vague symptoms like fatigue and hair loss, but blood tests are inconvenient or unavailable for many people. Our app lets users type symptoms in plain English, like “I feel tired and am experiencing hair loss.” It diagnoses the likely vitamin deficiency and then feeds that result into an LLM to create a custom meal plan.

3.2 Formal Definition

We address a supervised multi-class classification problem over 6 deficiency disease categories (Anemia, Scurvy, Rickets, Night Blindness, Healthy, and Other) using 4,000 patient records. The goal is to learn a function $f(x) \rightarrow y$ where x is a feature vector of patient-reported symptoms, demographics, and lifestyle factors, and y is the predicted deficiency category. The predicted label is then passed to an LLM to generate a personalized meal plan, bridging symptom-based diagnosis and dietary intervention in a single pipeline. This bridges the gap between previous machine learning models that focus only on prediction and LLM-based nutrition tools that lack an initial diagnosis from user symptoms.

4 Proposed Approach

We plan to use XGBoost and LightGBM as our primary models to predict vitamin deficiency from symptoms, demographics, and lifestyle factors. We chose these because they typically give higher accuracy, train faster, and handle overfitting better than simpler models on health data like this. Random Forest and Logistic Regression serve as our baselines, with Random Forest providing a strong comparison point and Logistic Regression acting as a minimal floor so we can clearly measure the improvement our primary models achieve. Free-text user input will be parsed into structured features using an open-source LLM (Qwen3.5-14B-Instruct, but may change), and the prediction result will then be passed to the same LLM to generate a personalized meal plan.

Potential challenges include:

- Computational complexity
- Convergence issues
- Data preprocessing requirements

5 Evaluation Plan

Our main metric is macro-averaged F1, which weighs all 6 deficiency classes equally and handles class imbalance better than plain accuracy. We also report per-class precision and recall to see where the model struggles, ROC-AUC for overall performance, and a confusion matrix to understand which diseases get mixed up. We use a stratified 80/20 train-test split with 5-fold cross-validation so every class is represented proportionally in each fold. SHAP values tell us which symptoms matter most for each prediction.

5.1 Success Criteria

We consider the project successful if XGBoost or LightGBM achieves the highest macro F1 among all four models with consistent performance across all 5 cross-validation folds, demonstrating that gradient boosting methods outperform the baselines on this task.

6 Dataset Plan

We will use datasets with clearly specified characteristics:

- CSV structured format
- About 4k records
- 35+ variable features
- Training/validation/test splits

Labeled data from public datasets to feed into a model for training purposes. This will be formatted in a .csv sheet style with columns of data. There will be categorical and numerical data since there will be symptoms that the user is required to enter. These symptoms will be given to the machine learning model. Data will be obtained from public datasets like Kaggle and be formatted by numerical (like age and percentage) and categorical (a minimum of 10 columns of symptoms).

7 Expected Outcomes

By the end of the semester, we expect to produce:

A machine learning model with a minimum accuracy of 90 percent that can take inputted data of symptoms and output a person's vitamin deficiency to help a person improve their health.

Possible deliverables include a developed model, tracked accuracy metrics, and feature engineering techniques.

8 Team Responsibilities

Each team member will contribute to distinct project components:

- Afeef Ali — Dataset Collection, Training Model
- Aarav Bafna — Model Research, Feature Engineering
- Tiffany Widjaja — Metrics Research, Feature Engineering
- Sanjana Gollu — Feature Engineering Research, Model Evaluation
- Natalie Rogers — Dataset Collection, Training Model
- All members — Writing and presentation

9 Timeline

Week 2/22-2/28	Dataset Collection and Model Research
Week 3/1-3/6	Preliminary Model Training Milestone 50% minimum accuracy
Week 3/15-3/21	Feature Engineering Milestone 75% minimum accuracy
Week 3/22-3/28	Feature Engineering Milestone 90% minimum accuracy
Weeks 3/29-4/4	Feature Engineering Milestone 90%+ accuracy and Metrics Evaluation
Weeks 4/5-4/14	Final Report and Presentation

10 Conclusion

This proposal outlines a structured plan for developing and evaluating a computational solution to a clearly defined problem. By combining insights from prior work with new methodological ideas, we aim to produce a robust and well-validated system. The project emphasizes clarity, reproducibility, and measurable progress.

References

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